

# Zhuolin Qu

## Curriculum Vitae

University of Texas at San Antonio  
Department of Mathematics  
Assistant Professor  
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### Education

- 2011–2016 **Doctor of Philosophy, Applied Mathematics, Tulane University.**  
Advisor: Alexander Kurganov
- Master of Science, Statistics, Tulane University.**  
Advisor: Michelle Lacey
- 2007–2011 **Bachelor of Science, Mathematics and Computational Science, University of Science and Technology of China.**  
Advisor: Mengping Zhang

### Professional Employment History

- 2020–present **Assistant Professor, Department of Mathematics, University of Texas at San Antonio.**
- 2016–2020 **Postdoctoral Fellow, Department of Mathematics, Tulane University.**  
Mentor: James (Mac) Hyman
- 2017–2019 **Visiting Scholar, Los Alamos National Laboratory.**  
(Summer) Theoretical Biology and Biophysics (T-6), Host: Benjamin McMahon
- 2015 **Summer Intern, Los Alamos National Laboratory.**  
Computational Earth Science (EES-16), Mentor: Carl Gable, Nataliia Makedonska
- 2012, 2013 **Visiting Fellow, Shanghai Jiao Tong University, China.**  
(Summer) Institute of Natural Sciences

### Research Interests

Mathematical and Computational Biology, Infectious Diseases Modeling, Population Dynamics, Numerical Methods for Nonlinear PDEs, Scientific Computing

### Selected Grants

- 2023-2026 **NSF Research Grant DMS-2316242**, “*LEAPS-MPS: Development of Novel Multistage Models for Wolbachia-Based Strategies to Control Mosquito-Borne diseases*”, Role: Sole PI, \$246,620.
- 2024-2025 **MAA Tensor Women & Mathematics**, “*Empowering Excellence: Women in Mathematics Workshop in San Antonio*”, Role: PI, \$5000.
- 2023-2024 **AWM-NSF Mathematics Mentoring Travel Grants**, “*Modeling Wolbachia-based interventions for malaria control*”, Role: Sole PI, \$5000.
- 2023 **AMS-Simons Research Enhancement Grant for Primarily Undergraduate Institution (PUI) Faculty**, Role: Sole PI, *Declined*.

- 2022 **Convergence Accelerator Team (CAT) Award**, “Developing Methodologies for Spatial and Demographic Heterogeneity in Malaria Immune Dynamics”, NSF-Simons Center for Multiscale Cell Fate Research, Role: PI, \$10,310.
- 2020 **MIDAS COVID-19 Urgent Grant - Supplemental Funding**, “Optimizing COVID-19 Awareness and Testing Strategy”, MIDASUGP2020-2, \$9,999.  
Role: Co-Investigator (25% contribution)

## Publications and Preprints

(\*denotes authors in alphabetical order, <sup>†</sup>denotes corresponding author)

- Published
1. **Qu, Zhuolin<sup>†</sup>** and Childs, L., Assessing the impact of the *Wolbachia*-based control of malaria, *Mathematical Biosciences*, 2025, 387; 109466.
  2. **Qu, Zhuolin<sup>\*</sup>**, Patterson, D.<sup>\*</sup>, Zhao, L., Ponce, J., Edholm, C., Prosper, O., and Childs, L.<sup>†</sup>, Mathematical modeling of malaria vaccination with seasonality and immune feedback. *PLoS Computational Biology*, 2025, 21(5); e1012988. <sup>\*</sup>denotes joint first authorship.
  3. **Qu, Zhuolin<sup>\*</sup>**, Patterson, D.<sup>\*</sup>, Childs, L.<sup>†</sup>, Edholm, C., Ponce, J., Prosper, O., and Zhao, L. Modeling immunity to malaria with an age-structured PDE framework. *SIAM Journal on Applied Mathematics*, 2023, 83(3); 1098 - 1125. <sup>\*</sup>denotes joint first authorship.
  4. Florez, D. A., Young, A. J., Bernabe, K., Hyman, J. M., and **Qu, Zhuolin<sup>†</sup>**. Modeling sustained transmission of *Wolbachia* among *Anopheles* mosquitoes: implications for malaria control in Haiti. *Tropical Medicine and Infectious Disease*, 2023, 8(3), 162.
  5. **Qu, Zhuolin<sup>†</sup>**, Wu, T., Hyman, J. M. Modeling spatial waves of *Wolbachia* invasion for controlling mosquito-borne diseases. *SIAM Journal on Applied Mathematics*, 2022, 82(6); 1903-1929.
  6. Kurganov, A.<sup>†</sup>, **Qu, Zhuolin<sup>\*</sup>**, and Wu, T. Well-balanced positivity preserving adaptive moving mesh central-upwind schemes for the Saint-Venant system. *ESAIM: Mathematical Modelling and Numerical Analysis*, 2022, 56(4); 1327-1360.
  7. Gulbudak, H.<sup>†</sup>, **Qu, Zhuolin**, Milner, F., and Tuncer, N., Sensitivity analysis in an immuno-epidemiological vector-host model. *Bulletin of Mathematical Biology*, 2022, 84; 27.
  8. Stoecker, C.<sup>†</sup>, Monette, A., **Qu, Zhuolin**, Schmidt, N., Craig-Kuh, M. C., and Kissinger, P. Cost effectiveness of Check It: a novel community-based chlamydia screening and expedited treatment program for young black men. *Clinical Infectious Diseases*, 2021; ciab818.
  9. Azizi, A.<sup>†</sup>, Dewar, J., **Qu, Zhuolin**, and Hyman, J. M. Using an agent-based sexual-network model to analyze the impact of mitigation efforts for controlling chlamydia. *Epidemics*, 2021, 35; 100456.
  10. **Qu, Zhuolin**, Azizi, A., Schmidt, N., Craig-Kuh, M. C., Stoecker, C., Hyman, J. M., and Kissinger, P.<sup>†</sup> Effect of screening young men for chlamydia trachomatis on the rates among women: a network modelling study for high-prevalence communities. *BMJ Open*, 2021, 11; e040789.
  11. **Qu, Zhuolin<sup>†</sup>**, McMahon, B. H., Perkins, D. J., and Hyman, J. M. Staged progression epidemic models for the transmission of invasive nontyphoidal Salmonella (iNTS) with treatment. *Mathematical Biosciences and Engineering*, 2021, 18(2): 1529-1549.

12. Azizi, A.<sup>†</sup>, **Qu, Zhuolin**, Lewis, B., and Hyman, J. M. Generating a heterosexual bipartite network embedded in social network. *Applied Network Science*, 2021, 6, 30.
13. Kurganov, A.<sup>†</sup>, **Qu, Zhuolin**<sup>\*</sup>, Rozanova, O. S., and Wu, T. Adaptive moving mesh central-upwind schemes for hyperbolic system of PDEs: applications to compressible Euler equations and granular hydrodynamics, *Communications on Applied Mathematics and Computation*, 2021, 3, 445–479.
14. Thongsripong, P.<sup>†</sup>, **Qu, Zhuolin**, Yukich, J., Hyman J. M., and Wesson, D. An investigation of human-mosquito contact using surveys and its application in assessing dengue viral transmission risk. *Journal of Medical Entomology*, 2020, 57(6), 1942–1954.
15. **Qu, Zhuolin**<sup>†</sup> and Hyman, J. M. Generating a hierarchy of reduced models for a system of differential equations modeling the spread of *Wolbachia* in mosquitoes. *SIAM Journal on Applied Mathematics*, 2019, 79(5):1675-1699.
16. **Qu, Zhuolin**<sup>†</sup>, Xue, L., and Hyman, J. M. Modeling the transmission of *Wolbachia* in mosquitoes for controlling mosquito-borne diseases. *SIAM Journal on Applied Mathematics*, 2018, 78(2):826–852.
17. Cheng, Y. Z., Kurganov, A.<sup>†</sup>, **Qu, Zhuolin**<sup>\*</sup>, and Tang, T. Fast and stable explicit operator splitting methods for phase-field models. *Journal of Computational Physics*, 2015, 303:45-65.
18. Kao, C. Y., Kurganov, A.<sup>†</sup>, **Qu, Zhuolin**<sup>\*</sup>, and Wang, Y. A fast explicit operator splitting method for modified Buckley-Leverett equations. *Journal of Scientific Computing*, 2015, 64(3):837-857.
19. Chertock, A., Kurganov, A.<sup>†</sup>, **Qu, Zhuolin**<sup>\*</sup>, and Wu, T. Three-layer approximation of two-layer shallow water equations. *Mathematical Modelling and Analysis*, 2013, 18(5):675-693.

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## Selected Scholarly Presentations

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| Invited Talks | <ul style="list-style-type: none"> <li>○ <b>Mathematical Congress of the Americas 2025</b>, Mini-symposium talk, Miami, 07/2025</li> <li>○ <b>SMB Annual Meeting</b>, Mini-symposium talk, University of Alberta, Canada, 07/2025</li> <li>○ <b>SIAM Conference on Applications of Dynamical Systems</b>, Mini-symposium talk, Denver, 05/2025</li> <li>○ <b>CUNY Hunter College</b>, Math &amp; Stat Colloquium, virtual, 03/2025</li> <li>○ <b>Joint Mathematics Meetings 2025</b>, Mini-symposium talk, Seattle, 01/2025</li> </ul>                                                                                                                                                                          |
| 2024          | <ul style="list-style-type: none"> <li>○ <b>The 14th AIMS Conference</b>, Mini-symposium talk, Abu Dhabi, 12/2024</li> <li>○ <b>Eighth Singapore International Dengue Workshop</b>, invited lecture, virtual, 11/2024</li> <li>○ <b>University of Georgia</b>, Applied Math Seminar, virtual, 09/2024</li> <li>○ <b>University of West Indies, Mona, Jamaica</b>, Colloquium talk, virtual, 09/2024</li> <li>○ <b>AMS Fall Central Sectional Meeting</b>, Mini-symposium talk, University of Texas at San Antonio, 09/2024</li> <li>○ <b>SIAM Annual Meeting</b>, Mini-symposium talk, Spokane, Washington, 07/2024</li> <li>○ <b>SMB Annual Meeting</b>, Mini-symposium talk, Seoul, Korea, 07/2024</li> </ul> |

- **Gordon Research Conference**, Genetic Biocontrol, Castelldefels, Spain, 05/2024
- **AMS Spring Eastern Sectional Meeting**, Mini-symposium talk, Howard University, 04/2024
- **Joint Mathematics Meetings 2024**, Mini-symposium talk, San Francisco, 01/2024
- 2023 ○ **University of Maryland**, Brin Mathematics Research Center, Mathematics of Malaria Transmission Dynamics, 11/2023
- **SIAM Texas-Louisiana Sectional Meeting**, Mini-symposium talk, University of Louisiana at Lafayette, 11/2023
- **AWM Research Symposium**, Mini-symposium talk, Clark Atlanta University, 10/2023
- **SMB Annual Meeting**, Mini-symposium talk, Ohio State University, 07/2023
- 2022 ○ **Southern University of Science and Technology**, PDE Seminar, 11/2022
- **SIAM Texas-Louisiana Sectional Meeting**, Mini-symposium talk, University of Houston, 11/2022
- **SIAM Conference on Life Sciences**, Mini-symposium, Pittsburgh, PA, 07/2022
- **AWM Research Symposium**, Mini-symposium, Univ. of Minnesota, 06/2022
- **The Fields Institute** for Research in Mathematical Sciences, Colloquium on Mathematics for Public Health, 04/2022
- **Joint Mathematics Meetings**, AWM special session, 04/2022
- **Joint Mathematics Meetings**, Mini-symposium talk, 04/2022
- **Oregon State University**, Mathematical Biology Seminar, 03/2022
- **Virginia Tech**, Mathematical Biology Seminar, 02/2022
- 2021 ○ **Middle Tennessee State University**, Mathematical Biology Seminar, 11/2021
- **SIAM Texas-Louisiana Sectional Meeting**, Mini-symposium talk, University of Texas Rio Grande Valley, South Padre Island, 11/2021
- **Kennesaw State University**, Mathematical Biology Seminar, 10/2021
- **Tulane University**, Applied Math Seminar, 10/2021
- **SMB Annual Meeting**, Mini-symposium talk, 06/2021
- **Utah State University**, Applied Math Seminar, 04/2021
- **Texas Tech University**, Biomathematics Seminar, 03/2021
- **Tulane University**, Guest lecture on “Epidemic Modeling Case Studies”, 02/2021
- **Joint Mathematics Meetings**, Mini-symposium talk, 01/2021
- 2020 ○ **Tulane University**, Guest lecture on “Modeling the Spread of Infectious Diseases”, 10/2020
- **SIAM Texas-Louisiana Sectional Meeting**, Mini-symposium talk, 10/2020
- Contributed ○ **9th Coastal Bend Mathematics and Statistics Conference (CBMSC)**,  
Talks Texas A&M International University, 04/2025
- **International Conference on Mathematical Modeling and Analysis of Populations in Biological Systems (ICMA VIII)**, University of Louisiana at Lafayette, 10/2022

- **SIAM Annual Meeting 2021**, virtual, 07/2021
- **Mathematics Research Communities**, virtual, 06/2021
- **Biomathematics and Ecology: Education and Research (BEER)**, virtual, 11/2020
- **CDC STD Conference 2020**, Atlanta, GA, virtual, 09/2020
- **Biology and Medicine through Mathematics (BAMM!)**, Virginia Commonwealth University, 05/2020 (abstract accepted, canceled due to COVID-19)
- **Scientific Computing around Louisiana**, Tulane University, 02/2019
- **Biomathematics and Ecology: Education and Research**, Arizona State University, 10/2018
- **SIAM Annual Meeting 2018**, Portland, OR, 07/2018
- **Scientific Computing around Louisiana**, Louisiana State University, 02/2018
- **SIAM Annual Meeting 2017**, Pittsburgh, PA, 07/2017
- **Scientific Computing around Louisiana**, Tulane University, 03/2017

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## Teaching Experience

Instructor University of Texas at San Antonio

- of Record
- MAT 6983: Master's thesis (graduate): Spring 2025, Fall 2025
  - MAT 5983/4953: Mathematical Biology (graduate): Spring 2024
  - MAT 5973: Directed Research (graduate): Summer 2023, Spring 2024, Fall 2024
  - MAT 5603: Numerical Analysis (graduate): Spring 2023
  - MAT 3633: Numerical Analysis, Fall 2021, Spring 2023, Fall 2023, Spring 2024, Fall 2024
  - MAT 3613: Differential Equations I: Fall 2022, Summer 2023, Fall 2023, Fall 2024, Spring 2025
  - MAT 2233: Linear Algebra, Spring 2022, Fall 2022 (online), Spring 2025
  - MAT 1193: Calculus for Biosciences, Fall 2020, Spring 2021(online)

Tulane University

- MATH 758: Scientific Computation III (graduate), Spring 2018, Spring 2019
- MATH 731: Applied Mathematics (graduate), Spring 2020
- MATH 221: Calculus III, Fall 2016
- MATH 122: Calculus II, Fall 2013
- MATH 116: Long Calculus II, Spring 2014

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## Mentoring Experience

- Doctoral Dissertation
- Assist in mentoring doctoral dissertation, Mathematics student, on modeling Wolbachia in mosquitoes with fractional diffusion (Tulane, 2023-2024)
  - Assist in mentoring doctoral dissertation, Mathematics student, on modeling epidemics with distribution parameters, committee member for Dissertation defense (Tulane, 2017-2019)
  - Assist in mentoring doctoral dissertation, Biostatistics student, on characterizing the spread of epidemics over networks (Tulane, 2018-2020)

- Assist in mentoring doctoral dissertation, Public Health student, on quantifying human-mosquito contact rate, paper published (Tulane, 2017-2018)
- Master Thesis
  - Joshua Lopez, Applied Mathematics, Thesis title: Assessing the Impact of Seasonality on Wolbachia Controls for Mosquito-Borne Diseases” (UTSA, 2023-2025)
  - Assist in mentoring master thesis project, Computational Science student, on modeling chikungunya disease and quantifying uncertainty (Tulane, 2018-2019)
  - Assist in mentoring master project, Computational Science student, on modeling the Chagas disease in the New Orleans area and estimating human risk. (Tulane, 2018-2020)
- Honor Thesis
  - Assist in mentoring undergraduate honors thesis, Neuroscience student, on modeling Tuberculosis progression with treatment, Senior Scholar Award for undergraduate (Tulane, 2018-2019)
  - Assist in mentoring undergraduate honors thesis, Mathematics student, on characterizing the stochastic spread of epidemics on networks (Tulane, 2018-2020)
- Directed Research
  - Behavior-driven socio-biological modeling of mosquito-borne diseases, one master student (UTSA, Spring 2025)
  - Modeling genetic-modified mosquitoes for disease controls, one master’s and one undergraduate student (UTSA, Fall 2024)
  - Introduction to mathematics epidemiology, two master students in Applied Mathematics (UTSA, Spring 2023)
  - Spatial models for Wolbachia-based mosquito control, one master student in Applied Mathematics (UTSA, Spring, Summer 2024)
- Course project
  - Assist in mentoring topic course project, on modeling Wolbachia transmission in Anopheles mosquitoes for Malaria control (Tulane, Spring 2021 - Spring 2023)
  - Assist in mentoring topic course project, on modeling Gonorrhea dynamics and interventions on networks (Tulane, Spring 2021)
- Conference Mentor
  - Society of Mathematical Biology annual meeting 2017, 2020, 2023, 2025
  - SIAM TX-LA Meeting 2022, 2023, 2024
- External Committee
  - PhD Committee, Daniela Florez, Mathematics, Tulane University, Spring 2022 - Spring 2024
  - PhD Dissertation Committee, Christian Geneus, Biostatistics, Tulane University, Spring 2020
  - Master Dissertation Committee, Harley Hanes, Center for Computational Science, Tulane University, Spring 2020
  - PhD Dissertation Committee, Li Guan, Mathematics, Tulane University, Fall 2019
  - Undergraduate Honors Thesis Prospectus, Rhea Kataria, Mathematics, Tulane University, Fall 2019
  - PhD Qualifying Exam Committee, Scientific Computing, Tulane University, Fall 2018

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## Service Activities

- Department
  - Scholarship and Awards Committee, Department of Mathematics, UTSA, Fall 2023 - present



- Technology Committee, Department of Mathematics, UTSA, Fall 2023 - present
- UTSA Day, Fall 2023, Spring 2025
- Search Committee, Mathematical Biology & Bioinformatics, UTSA, 2022 - 2023
- Admission Committee for Applied and Industrial Math Master (AIM) Program, Department of Mathematics, UTSA, 2020 - 2023
- Website Committee, Department of Mathematics, UTSA, 2020 - 2023
- College ◦ Search Committee, Dean for College of Sciences, UTSA, 2024
- Journal Journal of Biological Systems, Journal of Biological Dynamics, PLoS ONE, Journal
- Reviewer of Theoretical Biology, Mathematical Biosciences, Letters in Biomathematics, SIAM Journal on Applied Mathematics, PLOS Neglected Tropical Diseases
- Review Editor Frontiers in Tropical Diseases
- Guest Editor Mathematical Biosciences and Engineering, Special Issue: “Machine Learning, Mathematical and Statistical Modeling for Systems Biology”, 07/2022
- Tropical Medicine and Infectious Diseases, Special Issue: “Advancing mathematical models of mosquito-borne diseases”, 03/2023
- Organizer ◦ AMS Fall Central Sectional Meeting, mini-symposium, “Mathematics of Infectious Disease Emergence, Spread, and Control”, 09/2024
- “Empowering Excellence: Women in Mathematics Workshop in San Antonio”, University of Texas at San Antonio, 09/2024
- SIAM Annual Meeting, mini-symposium, “Mac’s Celebration of his 75th trip around the Sun”, 07/2024
- SIAM conference on Life Sciences, mini-symposium, “Epidemiological modeling across scales”, 06/2024
- SMB Annual 2023, mini-symposium, “Climate and vector-borne disease: insights from mathematical modeling”, 07/2023
- Joint Mathematics Meetings 2023, Special Session, “Advances in Modeling Mosquito-borne Disease Dynamics and Control Methods”, 01/2023
- SIAM Texas-Louisiana Meeting, mini-symposium, “Mathematical Modeling for Biological Dynamics”, 11/2022
- AWM Research Symposium, “Recent Developments in Ecological and Epidemiological Modeling”, University of Minnesota, 06/2022
- Joint Mathematics Meetings 2022, Special Session, “Dynamics of Infectious Diseases Across Multiple Scales”, 04/2022
- SIAM Texas-Louisiana Meeting, mini-symposium, “Mathematical and computational models for understanding emerging epidemics and evaluating intervention strategies”, 10/2020
- New Orleans workshop on Modeling the Spread of Infectious Diseases, Tulane University, Spring 2019 & Spring 2020
- Clifford Lectures, Co-organizer, Tulane University, 2017
- Reviewer ◦ NSF Panel reviewer, 06/2022
- SMB Annual Meeting, Reviewer for abstracts and Landahl grants, 03/2023, 04/2025
- MIDAS, Reviewer for conference, 09/2024

- Banff International Research Station (BIRS), Reviewer for proposals, 10/2024
- Media Coverage ○ **Math Horizons**, Evelyn J. Lamb, “Fighting an Epidemic with an Epidemic”, *Math Horizons*, 2019, 26:4, 22-23, DOI: 10.1080/10724117.2019.1574148
- **Forbes Magazine**, Innovation-Science, “Math-Based Mosquito Control To Prevent Human Diseases”, January 14th, 2019
- **Los Alamos Monitor**, “Solving epidemics with math”, October 10th, 2018
- **The Times-Picayune**, “Tulane researchers use math to contain the spread of mosquito-borne illnesses”, August 30th, 2018
- **SIAM News**, Research Nugget on “Sustained bacterial outbreak in mosquitoes limits spread of life-threatening diseases”, March 20th, 2018
- Poster Judge ○ SMB Poster Session, Edmonton, 2025
- Graduate Student Appreciation Week Poster Symposium, UTSA, 2025
- UT System Postdoctoral Symposium, UTSA, 2024
- School of Science and Engineering Research Day, Tulane University, 2018, 2019
- SIAM Texas-Louisiana Sectional Meeting, 2019
- Others ○ **MARC Mentor**, Maximizing Access to Research Careers (MARC) program, 2022
- **AWM and Math for America**, Student Essay Contest, Interviewee, 2022-2024
- **Newcomb Fellow**, Newcomb College Institute, 2016–2020, voluntary association of faculty from all undergraduate colleges to support women’s leadership/gender equity and foster faculty-student interaction and research